LUIS SERGE OLIVIER GERALDA TUTAB

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linkedin.com/in/luisgeralda | github.com/LuisGeralda | Portfolio
U.S. Permanent Resident (Green Card Holder)

Summary

5G/LTE RAN engineer with 5+ years of experience integrating eNB/gNB and core networks, validating radio and UE interoperability, and troubleshooting 3GPP protocol behavior. Combines ownership of end-to-end OTA test networks with Python software development for test automation, configuration analysis, and KPI reporting. Experienced in NSA/SA, ENDC/NRDC, wideband and subband beamforming, and SRS. Targeting RAN system validation and verification for terrestrial and non-terrestrial networks (NTN).

Technical Skills

Wireless/RAN: 5G NR NSA/SA, LTE, sub-6 GHz/mmWave, ENDC/NRDC, TDD/FDD, LAA (B46), CBRS (B48), mMIMO, carrier aggregation, wideband/subband beamforming, sounding reference signals (SRS)

Validation: Functional, regression, performance, interoperability, mobility, retainability, and stability testing; OTA test networks, test plans, root-cause analysis, RAN configuration and optimization

Protocols/Core: PHY, MAC, RLC, PDCP, RRC, NAS; S1AP, X2AP, NGAP, XnAP; EPC (MME, S-GW, P-GW, HSS), 5G Core (AMF, SMF, UPF), IMS, SIP, DIAMETER, VoLTE/VoNR, IPsec

Optimization: Handover thresholds, offsets, hysteresis, measurement events (A1–B2), reselection/redirection, DRX/eDRX, QoS (5QI), UL TCP Boost; NB-IoT, CAT-M

Diagnostics: QXDM, QCAT, Wireshark/TShark, XCAL/XCAP, PCAT, BTS counters, MAC TTI traces, ADB, iOS sysdiagnose

Software/Automation: Python, pandas, lxml, openpyxl, Bash, Linux, terminal/command line, Windows Task Scheduler; XML parsing, Excel reporting, CLI tools

O-RAN/Open Source: O-RAN fundamentals (eCPRI, F1, E2, A1, O1), srsRAN, Open5GS, OSMOCOM, SCAT; MATLAB 5G Toolbox familiarity

AI Development Tools: Claude Code, Cursor, GitHub Copilot, Codex

Experience

Nokia North America (F2G Solutions) Apr 2025 - Present
5G Technical Customer Advocate Client: Apple Inc.

- Support Apple test-network configuration and RAN optimization requests, investigating network parameter inconsistencies and analyzing implementation behavior against 3GPP specifications to guide troubleshooting.
- Independently led development of a Python automation framework for weekly LTE/NR cell inventories across labs in the US and Europe; parsed Nokia SCF/XML managed objects with lxml and generated Excel reports using pandas and openpyxl.
- Built configuration rules for intra/inter-frequency handovers, inter-RAT mobility, reselection, redirection, and carrier aggregation; checked feature flags, measurement settings, neighbor relations, and target-cell dependencies to identify blocked or incomplete configurations.
- Implemented comparisons of XML configuration snapshots and Excel inventories to expose added, removed, and changed parameters and cell relationships, supporting repeatable lab configuration reviews.

Corning Optical Communication (ex Spidercloud Wireless) Aug 2023 - Mar 2025
5G System Validation Engineer Milpitas, CA

- Owned and maintained four end-to-end 4G/5G OTA test networks used to evaluate performance, mobility, retainability, and stability across radio access, core network, and UE components.
- Led qualification of N3RU and M3L 5G radio units across 50+ indoor/outdoor coverage scenarios; developed system-level test plans, feature scripts, and troubleshooting playbooks, reducing qualification time by 25%.
- Integrated and optimized 5G NSA test networks using Druid Raemis 5G Core and Aricent 4G Core; refined handover thresholds and scheduling parameters to reduce call drops by 20%.
- Managed a 50-device Android/iOS fleet for real-world OTA testing; developed Python, Bash, ADB, and Apple Shortcuts workflows to automate UE interactions and reduce manual testing by 40%.

Experience continued

Corning Optical Communication (ex Spidercloud Wireless) Sep 2022 - Aug 2023
5G QA Test Engineer Milpitas, CA

- Executed regression, functional, and performance testing for 5G sub-6 GHz/mmWave solutions; monitored RSRP, RSRQ, and throughput KPIs to identify 15+ critical defects before software/hardware release.
- Developed Python and Bash scripts to parse QXDM, QCAT, and Wireshark logs, accelerating root-cause analysis by 30% and supporting investigation of RF and protocol behavior with development teams.
- Collaborated with product managers and R&D to refine feature specifications, reproduce RF anomalies, and verify corrective changes through follow-up testing.

Nokia North America (F2G Solutions) Apr 2021 - Sep 2022
4G & 5G System Engineer - Interoperability Dallas, TX

- Planned and executed 300+ cases from a 3,000+ test library for T-Mobile, AT&T, and Verizon across low, mid, and high bands, covering registration, handovers, reselections, redirections, carrier aggregation, VoLTE/VoNR, CSFB, E911, TTI bundling, and MFBI.
- Validated LTE/5G smartphone interoperability with Qualcomm, Samsung, and MediaTek chipsets; analyzed UE logs and LTE–WCDMA/LTE–5G NSA behavior to isolate multi-RAT connectivity issues.
- Commissioned Nokia radio modules (including AHIB, ARZB, and AZRA) and baseband units (ABIA, ASIA, ASIK, ABIL) to support new frequency bands and capacity expansions.
- Maintained Nokia RAN software (SBTS20A to SBTS22R3) and IPsec gateway configurations; coordinated parallel testing to reduce lab downtime by 15%.
- Correlated QXDM/QCAT, PCAT, Wireshark, Syslog, MAC TTI traces, and BTS counters to isolate eNB/gNB and UE issues; provided on-site and remote troubleshooting support to test teams and customer labs.
- Tuned mobility thresholds, offsets, and hysteresis to improve handover behavior; integrated NB-IoT/CAT-M devices and verified VoLTE/VoNR voice continuity.
- Reduced new-engineer ramp-up time by 30% through training in 3GPP-based test documentation, log analysis, and device interoperability with major chipset vendors.

Personal Projects

DTLog Python, FastAPI, React, TShark, SCAT
https://github.com/F2G-Telco-Academy/DTLog

- Built a Python analyzer that decodes SCAT-format PCAP captures from commercial off-the-shelf (COTS) UEs through TShark, reconstructing 2G/3G/LTE procedures and reporting signaling failures with decoded cause codes.
- Implemented state-machine procedure detection and retransmission deduplication to prevent double-counting; pooled mobility, accessibility, and voice KPIs across captures for scenario evaluation through a FastAPI/React dashboard and Excel reports.

Cellular Analyzer Kotlin, Android, Jetpack Compose
https://github.com/LuisGeralda/CellularAnalyzer

- Developed a Kotlin/Jetpack Compose app using Android telephony APIs to collect device-exposed LTE/NR measurements (RSRP, RSRQ, SINR), PCI, cell identity, and frequency information alongside GPS location.
- Implemented foreground-service logging, local session storage, map visualization, and CSV export to correlate radio measurements with drive-test routes and support post-test coverage analysis.

LTE KPI K-means Clustering Python, K-means
https://github.com/LuisGeralda/LTE-KPI-Kmeans-Clustering

- Applied K-means to LTE RSRP, RSRQ, and throughput data to identify coverage and cell-edge performance gaps; recommended parameter changes that improved coverage uniformity by 15% in a proof of concept.

Education

International Technological University GPA: 4.0
M.S. in Software Engineering (AI Concentration) Santa Clara, CA

Kennesaw State University GPA: 3.81
B.S. in Electrical Engineering Technology Marietta, GA